EMC First Midterm — Exam Preparation Examples

Course: Electromagnetic Compatibility, NTUST
Format: Closed-book midterm
Source of truth: chap1.pdf – chap4.pdf

Key Formulas Likely to Be Tested

Chapter 1 — Foundations

$$T_D = \frac{\mathscr{L}}{v}$$

(chap1.pdf, p.16)

$$v = \frac{1}{\sqrt{\epsilon\mu}} = \frac{v_0}{\sqrt{\epsilon_r \mu_r}}, \quad v_0 = 3 \times 10^8 \text{ m/s}$$

(chap1.pdf, p.20)

$$k = \frac{\mathscr{L}}{\lambda} \quad \text{(electrical dimension)}$$

(chap1.pdf, p.21)

$$\text{Voltage gain (dB)} = 20 \log_{10}\left(\frac{V_{\text{out}}}{V_{\text{in}}}\right), \quad \text{Power gain (dB)} = 10 \log_{10}\left(\frac{P_{\text{out}}}{P_{\text{in}}}\right)$$

(chap1.pdf, p.25–26)

$$\text{Cable loss (dB)} = 8.686\, \alpha \mathscr{L}$$

(chap1.pdf, p.30)

$$\text{dB}\mu\text{V} = 107 + \text{dBm} \quad (50\ \Omega \text{ system})$$

(chap1.pdf, p.30)

Chapter 2 — Regulatory Limits

Distance normalization for radiated emission comparisons:

$$\Delta E_{\mathrm{dB}} = 20 \log_{10}\left(\frac{d_1}{d_2}\right)$$

(chap2.pdf, p.13)

Class A vs. Class B at same distance: Class A is ~10 dB less stringent. *(chap2.pdf, p.13)*

Chapter 3 — Signal Spectra

$$c_n = \frac{1}{T}\int_{t_1}^{t_1+T} x(t)\, e^{-jn\omega_0 t}\, dt$$

(chap3.pdf, p.8)

$$|c_n| = \frac{A\tau}{T}\left|\frac{\sin(\pi f \tau)}{\pi f \tau}\right| \quad \text{(rectangular pulse train)}$$

(chap3.pdf, p.12)

$$f_1 = \frac{1}{\pi\tau}, \quad f_2 = \frac{1}{\pi\tau_r}, \quad \mathrm{BW} = \frac{1}{\tau_r}$$

(chap3.pdf, p.34, 37)

$$G_x(f) = \frac{X_0^2}{4}\,\delta(f) + \frac{X_0^2 T}{4}\,\frac{\sin^2(\pi f T)}{(\pi f T)^2} \quad \text{(PCM-NRZ PSD)}$$

(chap3.pdf, p.52)

Chapter 4 — Transmission Lines

$$\frac{\partial V}{\partial z} = -l\,\frac{\partial I}{\partial t}, \quad \frac{\partial I}{\partial z} = -c\,\frac{\partial V}{\partial t}$$

(chap4.pdf, p.8–9)

$$Z_C = \sqrt{\frac{l}{c}}, \quad v = \frac{1}{\sqrt{lc}}, \quad lc = \mu\epsilon$$

(chap4.pdf, p.12, 28)

$$\Gamma_L = \frac{R_L - Z_C}{R_L + Z_C}, \quad \Gamma_S = \frac{R_S - Z_C}{R_S + Z_C}$$

(chap4.pdf, p.35, 37)

$$V(0,t) = \frac{Z_C}{R_S + Z_C} V_S(t) \quad \text{(initially launched voltage)}$$

(chap4.pdf, p.37)

$$\tau_r > 10\, T_D \quad \text{(transmission line negligible criterion)}$$

(chap4.pdf, p.59)

$$\hat{Z}_{\text{in}} = Z_C \frac{\hat{Z}_L + jZ_C \tan \beta \mathscr{L}}{Z_C + j\hat{Z}_L \tan \beta \mathscr{L}}$$

(chap4.pdf, p.76)

$$\text{VSWR} = \frac{1 + |\hat{\Gamma}_L|}{1 - |\hat{\Gamma}_L|}$$

(chap4.pdf, p.77)

$$\hat{\gamma} = \sqrt{(r + j\omega l)(g + j\omega c)} = \alpha + j\beta$$

(chap4.pdf, p.81)

$$\delta = \frac{1}{\sqrt{\pi f \mu_0 \sigma}} \quad \text{(skin depth)}$$

(chap4.pdf, p.82)

$$\tan\theta = \frac{\epsilon_r''}{\epsilon_r'}, \quad g = \omega c \tan\theta$$

(chap4.pdf, p.87)

$$\alpha \approx \frac{1}{2}\left(\frac{r}{Z_C} + gZ_C\right) \quad \text{(low-loss)}$$

(chap4.pdf, p.91)

Example Problems from Slides

Example 1 — dBμV / dBm Conversion *(chap1.pdf, p.30)*

Problem: A signal source delivers 0 dBm into a 50 Ω spectrum analyzer. What is the voltage level in dBμV?

Solution:

$$\text{dB}\mu\text{V} = 107 + \text{dBm} = 107 + 0 = 107 \text{ dB}\mu\text{V}$$

(chap1.pdf, p.30)

Example 2 — Cable Loss Calculation *(chap1.pdf, p.30)*

Problem: A coaxial cable has attenuation constant $\alpha = 0.05$ Np/m. Find the cable loss in dB for a cable of length $\mathscr{L} = 10$ m.

Solution:

$$\text{Cable loss} = 8.686 \times 0.05 \times 10 = 4.34 \text{ dB}$$

(chap1.pdf, p.30)

Example 3 — Electrical Dimension and Wavelength *(chap1.pdf, p.21)*

Problem: A PCB trace is 17.78 cm long. The substrate is epoxy glass with $\epsilon_r = 4.7$, $\mu_r = 1$. At $f = 1$ GHz, is the trace electrically short?

Solution:

$$v = \frac{v_0}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{4.7}} = 1.383 \times 10^8 \text{ m/s}$$

$$\lambda = \frac{v}{f} = \frac{1.383 \times 10^8}{10^9} = 0.138 \text{ m} = 13.8 \text{ cm}$$

$$k = \frac{\mathscr{L}}{\lambda} = \frac{17.78}{13.8} = 1.29$$

The trace is **electrically long** ($k > 1$) at 1 GHz — transmission-line analysis is required.

(chap1.pdf, p.21; chap2.pdf, p.37)

Example 4 — Normalizing Radiated Emission Distance *(chap2.pdf, p.13)*

Problem: The FCC Class A limit at 30–88 MHz is 39 dBμV/m measured at 10 m. What is the equivalent limit at 3 m?

Solution:

$$\Delta E = 20\log_{10}\left(\frac{10}{3}\right) = 20\log_{10}(3.33) = 20 \times 0.523 = 10.46 \text{ dB}$$

$$\text{Limit at 3 m} = 39 + 10.46 = 49.46 \text{ dB}\mu\text{V/m}$$

Compare to Class B at 3 m: 40 dBμV/m — so Class A is ~9.5 dB less stringent.

(chap2.pdf, p.13)

Example 5 — Trapezoidal Spectral Bounds *(chap3.pdf, p.34)*

Problem: A clock signal has amplitude $A = 3.3$ V, period $T = 10$ ns ($f_0 = 100$ MHz), pulse width $\tau = 5$ ns (50% duty cycle), and rise time $\tau_r = 1$ ns.

(a) Find the spectral bound breakpoints f_1 and f_2.
(b) Find the DC level and peak envelope level.
(c) Estimate the level at $f = 200$ MHz.

Solution:

(a) Breakpoints:

$$f_1 = \frac{1}{\pi\tau} = \frac{1}{\pi \times 5 \times 10^{-9}} = 63.7 \text{ MHz}$$

$$f_2 = \frac{1}{\pi\tau_r} = \frac{1}{\pi \times 1 \times 10^{-9}} = 318 \text{ MHz}$$

(chap3.pdf, p.34)

(b) Levels:

$$c_0 = A\frac{\tau}{T} = 3.3 \times 0.5 = 1.65 \text{ V}$$

Peak envelope: $2A\tau/T = 2 \times 1.65 = 3.3 \text{ V} \rightarrow 20\log_{10}(3.3) = 10.4$ dBV *(chap3.pdf, p.30)*

(c) Estimate at $f = 200$ MHz (in the -20 dB/dec region, since $f_1 < 200$ MHz $< f_2$):

$$\text{Level}_{200} = \text{Level}_{f_1} - 20\log_{10}\left(\frac{200}{63.7}\right) = 10.4 - 20\log_{10}(3.14) = 10.4 - 9.9 = 0.5 \text{ dBV}$$

(chap3.pdf, p.34–36)

Example 6 — Transmission Line Reflection *(chap4.pdf, p.38)*

Problem (based on Example 4.1): A transmission line has $Z_C = 50\,\Omega$, $v = 200$ m/μs, $\mathscr{L} = 400$ m. The source is a 30 V step with $R_S = 0\,\Omega$ and the load is $R_L = 100\,\Omega$.

(a) Find T_D, Γ_L, and Γ_S.
(b) Find the initially launched voltage.
(c) Find the load voltage at $t = T_D^+$ and $t = 3T_D^+$.

Solution:

(a)

$$T_D = \frac{400}{200 \times 10^6} = 2\,\mu\text{s}$$

$$\Gamma_L = \frac{100 - 50}{100 + 50} = \frac{50}{150} = \frac{1}{3}$$

$$\Gamma_S = \frac{0 - 50}{0 + 50} = -1$$

(chap4.pdf, p.35, 37)

(b) Initial voltage:

$$V^+ = \frac{Z_C}{R_S + Z_C} V_S = \frac{50}{0 + 50} \times 30 = 30 \text{ V}$$

(chap4.pdf, p.37)

(c) Load voltage:

At $t = T_D^+$ (first arrival at load):

$$V(\mathscr{L}) = V^+(1 + \Gamma_L) = 30\left(1 + \frac{1}{3}\right) = 40 \text{ V}$$

Reflected wave $V^- = \Gamma_L V^+ = 10$ V travels back, reaches source at $t = 2T_D$. At source: re-reflected by $\Gamma_S = -1$, so $V^{++} = -10$ V.

At $t = 3T_D^+$ (second arrival at load):

$$V(\mathscr{L}) = 40 + (-10)(1 + \Gamma_L) = 40 - 10 \times \frac{4}{3} = 40 - 13.3 = 26.7 \text{ V}$$

(chap4.pdf, p.38, 43)

Steady state: $V_{SS} = V_S \times R_L/(R_S + R_L) = 30 \times 100/100 = 30$ V.

Example 7 — Is the Transmission Line Important? ***(chap4.pdf, p.59)***

Problem: A PCB trace has $\mathscr{L} = 10$ cm, $\epsilon_r = 4.7$ (microstrip). A CMOS gate drives it with rise time $\tau_r = 1$ ns. Does the trace behave as a transmission line?

Solution:

$$v = \frac{v_0}{\sqrt{\epsilon_r'}} \approx \frac{3 \times 10^8}{\sqrt{4.7}} = 1.38 \times 10^8 \text{ m/s}$$

$$T_D = \frac{0.10}{1.38 \times 10^8} = 0.72 \text{ ns}$$

Check criterion: $\tau_r = 1$ ns vs. $10T_D = 7.2$ ns.

$$\tau_r < 10T_D \quad (1 \text{ ns} < 7.2 \text{ ns})$$

The line matters — transmission-line analysis required. Reflections and overshoot will occur.

(chap4.pdf, p.59)

Example 8 — VSWR Calculation ***(chap4.pdf, p.77)***

Problem: A 50 Ω transmission line is terminated with $R_L = 150\,\Omega$. Find Γ_L and VSWR.

Solution:

$$\Gamma_L = \frac{150 - 50}{150 + 50} = \frac{100}{200} = 0.5$$

$$\text{VSWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = \frac{1.5}{0.5} = 3$$

(chap4.pdf, p.77)

Example 9 — Skin Depth *(chap4.pdf, p.82)*

Problem: Find the skin depth in copper ($\sigma = 5.8 \times 10^7$ S/m) at $f = 100$ MHz.

Solution:

$$\delta = \frac{1}{\sqrt{\pi f \mu_0 \sigma}} = \frac{1}{\sqrt{\pi \times 10^8 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7}}$$

$$= \frac{1}{\sqrt{\pi \times 10^8 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7}}$$

$$= \frac{1}{\sqrt{4\pi^2 \times 5.8 \times 10^8}} = \frac{1}{\sqrt{2.285 \times 10^{10}}} = \frac{1}{1.512 \times 10^5} \approx 6.6\,\mu\text{m}$$

(chap4.pdf, p.82)

Example 10 — Low-Loss Attenuation Constant *(chap4.pdf, p.91)*

Problem: A coaxial cable at 1 GHz has: $r = 0.1\,\Omega/\text{m}$, $g = 2 \times 10^{-4}$ S/m, $Z_C = 50\,\Omega$. Find α using the low-loss approximation. Then find the cable loss for $\mathscr{L} = 2$ m.

Solution (low-loss, verify $r \ll \omega l$ and $g \ll \omega c$ — assumed satisfied):

$$\alpha \approx \frac{1}{2}\left(\frac{r}{Z_C} + gZ_C\right) = \frac{1}{2}\left(\frac{0.1}{50} + 2 \times 10^{-4} \times 50\right)$$

$$= \frac{1}{2}(0.002 + 0.01) = \frac{0.012}{2} = 0.006 \text{ Np/m}$$

$$\text{Cable loss} = 8.686 \times 0.006 \times 2 = 0.104 \text{ dB}$$

(chap4.pdf, p.89–91)

Concepts Most Likely Tested on Closed-Book Exam

Topic	Key Concept	Where
EMC definition	Three criteria (cause/susceptible/self)	*chap1.pdf, p.4*
Three-element model	Source → Path → Receptor	*chap1.pdf, p.5*
Electrical dimension	$k = \mathscr{L}/\lambda$; lumped vs. distributed	*chap1.pdf, p.21*
dB conversions	20 log (V/I), 10 log (P); 6 dB = ratio of 2	*chap1.pdf, p.25–27*
Cable loss	$8.686\alpha\mathscr{L}$ dB	*chap1.pdf, p.30*
FCC Class A vs. B	Residential vs. commercial; B more stringent	*chap2.pdf, p.6*
Conducted emissions	LISN; 150 kHz–30 MHz	*chap2.pdf, p.7, 29*
Distance normalization	$20\log_{10}(d_1/d_2)$	*chap2.pdf, p.13*
Trapezoidal spectrum	Two-sinc product; three-region Bode	*chap3.pdf, p.29, 34*
Breakpoints	$f_1 = 1/(\pi\tau)$, $f_2 = 1/(\pi\tau_r)$	*chap3.pdf, p.34*
Bandwidth	$\mathrm{BW} = 1/\tau_r$	*chap3.pdf, p.37*
Rise time vs. emissions	Slower edges → less high-freq content	*chap3.pdf, p.35*
PCM-NRZ PSD	Delta at DC + sinc-squared continuous	*chap3.pdf, p.52*
Telegrapher's equations	Two coupled PDEs	*chap4.pdf, p.8–9*
$lc = \mu\epsilon$	Velocity formula; $v = 1/\sqrt{lc}$	*chap4.pdf, p.12*
Reflection coefficients	Γ_L and Γ_S; open = +1, short = -1	*chap4.pdf, p.35, 37*
Launched voltage	Voltage divider between R_S and Z_C	*chap4.pdf, p.37*
TL negligibility	$\tau_r > 10T_D$	*chap4.pdf, p.59*
Input impedance formula	$Z_{\text{in}} = Z_C \frac{Z_L + jZ_C \tan\beta\ell}{Z_C + jZ_L \tan\beta\ell}$	*chap4.pdf, p.76*
VSWR	$(1 + \lVert\Gamma\rVert)/(1 - \lVert\Gamma\rVert)$	*chap4.pdf, p.77*
Skin depth	$\delta = 1/\sqrt{\pi f \mu_0 \sigma}$; $\delta \propto 1/\sqrt{f}$	*chap4.pdf, p.82*
Loss tangent	$\tan\theta = \epsilon_r''/\epsilon_r'$; FR-4 ≈ 0.02	*chap4.pdf, p.87*
Low-loss α	$\frac{1}{2}(r/Z_C + gZ_C)$	*chap4.pdf, p.91*